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A New Indoor Environmental Quality Equation for Air-Conditioned Buildings

K.W. Mui[*†] and W.T. Chan[*]

Sustaining indoor environmental quality (IEQ) in high rise centrally air-conditioned buildings remains a difficult task of building managers because of the lack of a suitable instrument to measure the physical parameters of IEQ and to record the simultaneous occupant subjective responses. An IEQ logger is developed to facilitate the continuous monitoring of the state of IEQ. The performance of this logger has been validated by large-scale surveys in air-conditioned buildings in Hong Kong using the appropriate international standards and protocols.

The compactness of the logger made it versatile for use in the study of total IEQ. It renders the development of a total IEQ comfort model possible. This composite IEQ index attempts to integrate the total responses of the four fundamental indoor environmental qualifiers, namely, thermal, air quality, visual and aural comfort. The IEQ index expresses the dissatisfaction as can be predicted by measurements of the essential parameters. The concept of an IEQ index opens up a new approach to the design for acceptable indoor IEQ. The IEQ logger facilitates building managers in sustaining their targeted IEQ levels.

Introduction: An Integrated Approach to Environmental Comfort

In an indoor space, the ergonomic conditions can be described by a set of environmental qualifiers, namely, thermal comfort, indoor air quality, visual and aural conditions, furniture, working space, ease of movement, circulation, sight of windows, etc. Of these, thermal, indoor air, visual and aural comforts are sustained by the installed building services systems. Despite the intensive and extensive research into indoor environmental quality (IEQ), such quality in air-conditioned spaces remains a problem at building design, operation and maintenance levels. At design level, the individual qualifiers of IEQ, namely, the thermal comfort, indoor air quality, visual and aural comforts are often dealt with individually, disregarding the fact that occupants respond to the 'total' quality of their micro-environment. At the operation and maintenance level, it lacks a suitable instrument to evaluate the state of the overall indoor environment. In reality, if well serviced building is healthy to work in, but with unacceptable indoor environmental conditions, the occupants will manifest their protest in one or more forms of 'sick building syndrome'.

This paper proposes a new IEQ equation to compute an integrated IEQ index for more comprehensive control of the indoor environment embracing the four basic physical comfort qualifiers. An IEQ logger has been developed to facilitate the measurement of this index [1]. This logger can be used by the building managers for the monitoring of indoor environment.

To some extent the integrated response to total indoor environmental quality has been addressed [2,3]. Discomfort caused by the different physical qualities has been investigated. Clausen reported the relative discomfort caused by indoor air pollution, thermal load and noise [4]. It was found that when the operative temperature was in the range of 23 to 29°C, a 1°C change in operative temperature was found to have the same effect on comfort as a change in perceived air quality of 2.4 decipol, or a change in noise level of 3.9 dB. For levels of perceived air quality up to 10 decipol, a 1 decipol change in perceived air quality had the same effect on comfort as a change in noise level of 1.2 dB [4,5]. This result can be interpreted as that in general, people response more to a total sensation rather than to a specific indoor environmental qualifier. It opens up the opportunity to trade off the criteria for an optimized indoor environmental condition and energy effectiveness.

In an initial effort to develop an integrated IEQ index, the individual comfort qualifiers were investigated by the authors using an indoor en-

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vironmental quality survey. A large-scale in-office survey was conducted according to protocols based on international standards and accepted research practices. It involved extensive measurements of physical parameters and subjective surveys for the four basic environmental factors [6,7,8]. From this survey, a set of influencing parameters were identified.

Subsequently, an IEQ logger was designed with built in sensors for the measurements of these parameters, together with automatic logging of subjective responses, so that the various comfort indices could be calculated from the data collected. The performance of this logger was validated using the results from the large-scale survey [1]. The development of a new IEQ comfort equation is described in detail in this paper.

Methodology

Development of the IEQ Logger

There is no standard protocol available to assess the total indoor environmental quality of offices in air-conditioned buildings. The development of the IEQ logger was prompted by experience from the in-office survey. During 1992 and 1995, a comprehensive study of 16 offices in 13 high-rise buildings in Hong Kong was carried out by the Department of Building Services Engineering [6,7,8]. A comprehensive set of physical parameters was measured, together with the administration of a questionnaire to record subjective responses of the occupants, in order to compute the various comfort indices.

The overall complexity of such a scientific approach is outside the scope of assessments which can be undertaken by building operators. A simple and compact instrument capable of simultaneously measuring essential environmental parameters and subjective responses would be necessary for the non-specialist to ascertain the IEQ. In addition, with such an instrument, researchers can conveniently perform more extensive IEQ measurements in order to further understand the human responses to IEQ and thereby can better integrate the IEQ components.

An IEQ logger would be useful for routine measurements in offices given the following characteristics:

- small and compact size;
- records the essential physical parameters with sufficient accuracy;
- records occupant subjective responses to the indoor environment;
- the physical and subjective measurements can be used to derive other useful comfort indices; and
- the protocol requiring responses from building occupants is relatively simple and easily understood.

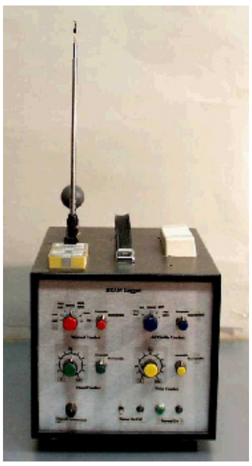
To make the IEQ logger small and compact, only the essential sensors are included, and only subjective responses of interest are recorded (Table 1). With the above considerations and criteria, the specification of the sensors selected was based on specifications in relevant standards [9].

Table 2 Sites surveyed

Building	System	Fresh Air System	Seasons	Types of Office	Location
A	CAV	Individual	Winter and Summer	Multi-tenant	North Point
B	FCU	Central	Winter	Open plan	Mong Kok
C	FCU	Central	Winter and Summer	Multi-tenant	Wan Chai
D	VAV	Central	Summer	Open plan	Wan Chai
E	CAV	Central	Winter	Open plan	Tsimshatsui
F	CAV	Central	Winter	Open plan	Central

Key: VAV – Variable Air Volume; CAV – Constant Air Volume; FCU – Fan Coil Unit

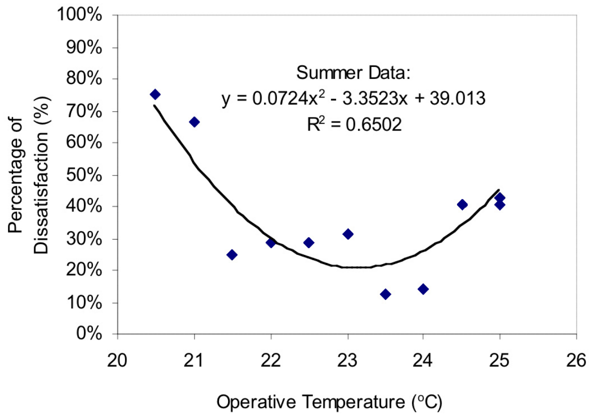
Table 1 IEQ logger and the Essential Set of Measured Parameters

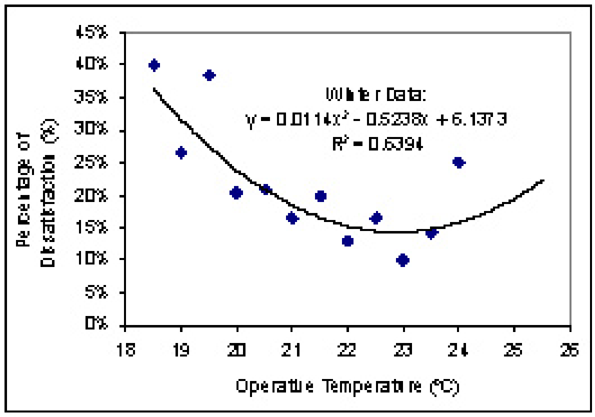


IEQ Qualifiers	Physical Parameters	Subjective Parameters
Thermal comfort	Air temperature	Thermal comfort votes
	Radiant temperature	Acceptability
	Globe temeprature	
Indoor air quality	Carbon dioxide	IAQ sensation vote
		Acceptability
Visual comfort	Horizontal illuminance	Visual sensation vote
		Acceptability
Aural comfort	Background noise level	Aural sensation vote
		Acceptability

Table 3. Summary of Thermal Comfort Quality Results

Season	Thermal comfort indices	HK-IEQ logger Performance [1]	Comprehensive study [6]
Summer	Regression equation	Mean vote = 0.3517 t_o – 8.1642	Mean vote = 0.3618 t_o – 8.5154
	Correlation coefficient	r^2=0.57	r^2=0.86
	Neutral temperature	23.2°C	23.5°C
	The operative temperature at minimum dissatisfaction (ASHRAE 55)	Between 22.5°C and 23°C	24°C
Winter	Regression equation	Mean vote = 0.2384 t_o - 5.3645	Mean vote = 0.2215 t_o – 4.7047
	Correlation coefficient	r^2=0.82	r^2=0.80
	Neutral temperature	22.2°C	21.5°C
	The operative temperature at minimum dissatisfaction	22°C	22°C





Vaildation Remarks: An extensive survey in offices in Hong Kong and showed that 60% of the over 1,000 work stations were on the cool and cold side with reference to ASHRAE 55 [6]. The neutral and preferred temperatures reported in this paper agree with those found in the extensive survey. It can be ascribed this as a 'social culture' rather than a normal response which is a consequence of misunderstanding of perceived thermal comfort. It is common to find people wearing jackets in the cool office environment in hot summer days.

Validation of the IEQ Logger

The field measurements were carried out in a summer and a winter. The information of the six buildings is shown in Table 2. The number of survey subjects was 422, with a sample size of 128 in three offices in the summer test and 294 in five offices in the winter test. The detail procedures of the validation of the IEQ logger was reported, the large scale in-office study data was used as reference for validation (Table 3 and 4) [1,6,7,8].

From the comparison, the IEQ logger has proven to conform to the results of the comprehensive study which was conducted with research grade measurement protocols of internationally accepted standards and research practices. The IEQ logger greatly reduces the effort of measurement in assessing indoor environmental quality. A useful IEQ index is then developed

Development of the IEQ Equation

Most previous studies of IEQ took account of only a few variables at one time, and failed to examine the interaction among different environmental components. Some researchers have considered the concept of total environmental quality and use multivariate regression analysis to indicate the workplace variable inducing the largest number of health symptoms, comfort or odour concerns, to establish a thermal comfort model based on field measurements [11,12,13]. The following is a simpler and logical way for the use by building managers.

First Stage Multiple Regression Equation for IEQ

Adopting a linear regression model, the Percentage of Dissatisfaction (PD) for IEQ can be said to be a linear function of the PDs for Thermal Comfort (TC), Indoor Air Quality (IAQ), Aural Comfort (AC), and Visual Comfort (VC). The general equation is:

$$PDIEQ = a_1 \times PDTC + a_2 \times PDIAQ + a_3 \times PDVC + a_4 \times PDAC \quad (1)$$

Where:

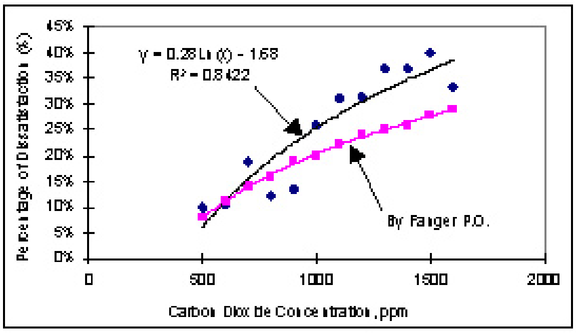
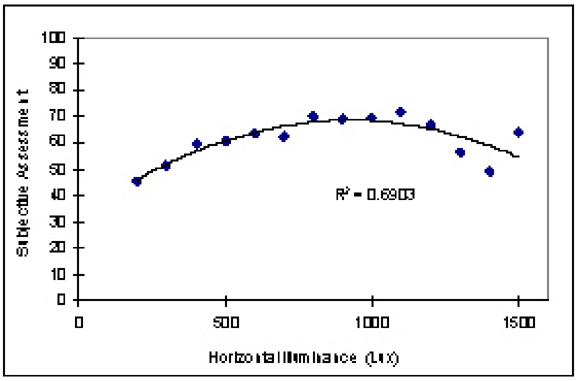
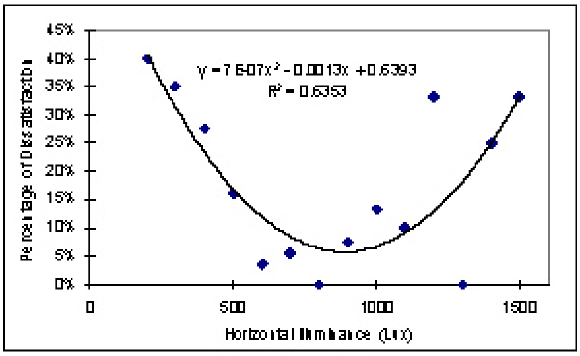
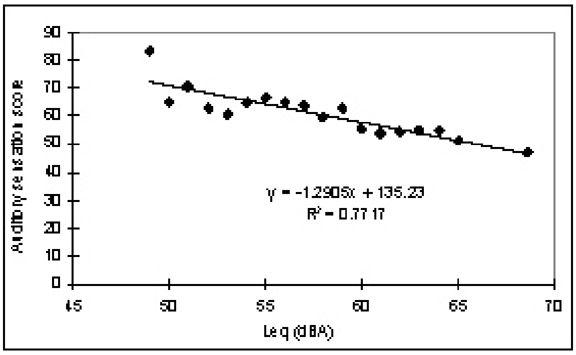
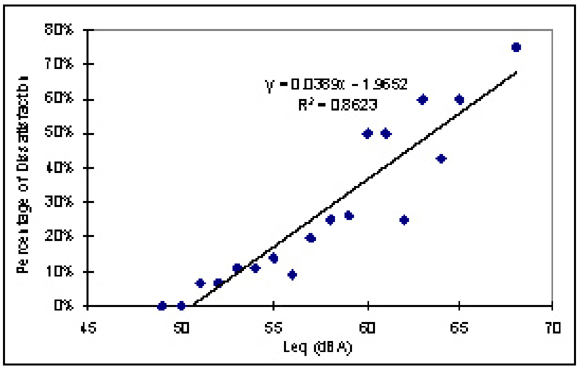
PDIEQ = Percentage of Dissatisfaction in indoor environmental quality

PDTC = Percentage of Dissatisfaction in thermal comfort

PDIAQ = Percentage of Dissatisfaction in indoor air quality

PDVC = Percentage of Dissatisfaction in visual comfort

Table 4 Validation of the IEQ meter for IAQ, Visual and Aural Comforts

IEQ Logger	Validation Remark
	Indoor Air Quality Comfort: The trend compares well with Fanger [10]. A higher percentage of dissatisfaction of 25% at 1,000 ppm is expected because there are more uncontrollable factors in real offices than in study chambers.
	Visual Comfort: Satisfaction and acceptability both peak at horizontal illuminance 800 to 1000 lux. The preferred values for visual comfort agree with those found in the extensive in-office survey over 2000 work stations [8].
	Visual Comfort: The percentage of dissatisfaction was lowest about the same range of horizontal illuminance at 5%.
	Aural comfort: For 50% score defined as acceptable and considered as 'neutral' (middle of quiet and nosy side), the neutral noise level was found at 66 dBA as compared to 53.3 dBA [8]. Trend behavior is the same in both cases.
	Aural comfort: The observed percentage of dissatisfaction was regressed with the L_{eq}(dBA). The occupants in this study showed zero dissatisfaction at all at a noise level of 51dBA (around NC 45), which is quite reasonable compared with the usual design criteria.

PDAC = Percentage of Dissatisfaction in aural comfort

a_1, a_2, a_3, a_4 are the coefficients for each of the comfort indices

Using the data obtained from the measurements with the IEQ logger, the multiple regressed IEQ equation was generated, as given in Table 5. The first stage IEQ equation was established as in Equation (2).

Table 5 First regression analysis for prediction of IEQ

	Coefficients	Standard Error	t Statistic	Lower 95%	Upper 95%
PDTC	0.44	0.12	3.53	0.15	0.72
PDIAQ	0.11	0.12	0.92	-0.16	0.38
PDVC	-0.23	0.33	-0.69	-0.98	0.53
PDAC	0.43	0.35	1.21	-0.38	1.24

$$PDIEQ = 0.44 \times PDTC + 0.11 \times PDIAQ - 0.23 \times PDVC + 0.43 \times PDAC \quad (2)$$

From the regressed IEQ Equation (2), it was found that the coefficient a_3 was negative, which is illogical. The reason is that range of illuminance on the working plane in the sample was quite narrow and is within the higher range of satisfaction, and hence give no significant contribution to the overall PD of the IEQ. Therefore, the PDVC was taken out of the model. The PDIEQ was regressed again with the other three PDs. The results were shown in Table 6 and the revised first stage IEQ equation is as shown in Equation (3).

Table 6. Second regression analysis for prediction of IEQ

	Coefficients	Standard Error	t Statistic	Lower 95%	Upper 95%
PDTC	0.42	0.12	3.57	0.15	0.69
PDIAQ	0.09	0.11	0.79	-0.16	0.34
PDAC	0.28	0.27	1.03	-0.33	0.88

$$PDIEQ = 0.42 \times PDTC + 0.09 \times PDIAQ + 0.28 \times PDAC \quad (3)$$

Equation (3) is significant in this study as the new IEQ equation because it relates the basic IEQ components together. Its significance lies in the fact that occupant response is an integration of all his/her microenvironment. In a new design, this equation becomes a new total PPD equation. When the PPDs are set as design targets, the PPDIEQ predicts the expectation for the total IEQ.

Looking at the coefficients of the regression equation, a few points can be noted:

- In most of cases, thermal comfort receives more complaints among the four basic environmental components. First of all, humans are very sensitive to air temperature, as judged from their variation of response within a small range of air temperature. Secondly, even in a typical office, the ranges of the other parameters such as the metabolic rate, clo values (clothing insulation), etc. can be significant;

- The small coefficient for IAQ seems to contradict the concern of IAQ in our buildings. It is due to the fact that IAQ is not as readily discernable. It is normally noticed only when any odour is objectionable. Under normal situations, an office is supposed to have no strong odour IAQ sources. Therefore, IAQ unacceptability will only spell out when the office is 'overcrowded' with respect to ventilation rate which generates intolerable level of human odours.

Surprisingly, the coefficient for PDAC is relatively high, and that of PDVC is insignificant. It is the fact that the lighting design target for illumination is much easier to attain. On the contrary, the prediction of the background noise is much more difficult and harder to achieve and should be given more attention.

Second Stage Multiple Regression Equation for IEQ

In the second stage of regressed IEQ equation, PDTC, PDIAQ, PDVC and PDAC were substituted by the corresponding comfort index equations, established by measurement. Equation (3) is modified to Equation (7).

According to the percentage of dissatisfaction on thermal comfort of the summer and winter data in this study:

$$PDTC = (0.021 \times t_o^2 - 0.919 \times t_o + 10.225) \times 100\% \quad (4)$$

Similarly,

$$PDIAQ = [0.28 \ln (CO_2\ ppm) - 1.68] \times 100\% \quad (5)$$

$$PDAC = [0.0389 \times dBA - 1.9652] \times 100\% \quad (6)$$

Substituting Equations (4), (5) and (6) in Equation (3), the overall IEQ equation is generated.

$$PDIEQ = [0.42 \times (0.021 \times t_o^2 - 0.919 \times t_o + 10.225) + 0.09 \times [0.28 \ln(CO_2) - 1.68] + 0.28 \times (0.0389 \times dBA - 1.9652)] \times 100\% \quad (7)$$

Equation (7) is the new IEQ Equation. *PDIEQ* is the new IEQ index and predicts the overall dissatisfaction of the indoor environmental quality when the fundamental physical parameters t_o, CO_2 and dBA are measured or set as design criteria. Of course, the application of this equation assumes an air-conditioned office with clean outdoor air, no appreciable indoor air pollutants, normal dress code for occupants and having satisfactory illumination. In fact, this equation uses t_o, CO_2 and dBA as surrogate indicators to represent the other parameters lumped together.

General Practice for Predicted Percentage of Dissatisfaction

The IEQ equation implicitly shows that it is impossible to have a 0% dissatisfaction. In optimising the resources of design, operation and maintenance, building developers can target IEQ index at 10%, 20% and 30% of dissatisfaction. Taking this approach, from the observation of the PPDs found in this study, together with the discussion from experienced professionals, the practice resulting to these three levels of dissatisfaction in the air-conditioning offices in Hong Kong can be generalised as follows:

i) Level 1 – 10% dissatisfaction of total IEQ.

It represents the highest grade of office environment which has an indoor environmental performance better than average. It requires more effort in design consideration; installation supervision; high precision testing, commissioning and fine tunning; user - oriented operation and predictive maintenance. The project team should realise that more resources are required during design, construction, choice of equipment and material, operation and maintenance. However, it gains the return of higher satisfaction rate, higher productivity, less down time of system and equipment, less recurrent costs in trouble shooting, remedial measures and replacement.

ii) Level 2 – 20% dissatisfaction of total IEQ.

It represents an average grade office. It can normally be achieved by traditional competence and ethical job when the appropriate standards for each of the basic IEQ components are complied with.

iii) Level 3 – 30% dissatisfaction of total IEQ.

This represents a below average grade office which is often the result of an investment plan based on speculation. In the building services industry, sick building syndrome is said to occur if 20 or more percent of the occupants complain about sick symptoms. The occupants may need to work out a scheme to reduce the loading to the system in order to reduce problems. There may be some saving in first cost and energy. The low return of satisfaction may prove to be worthless. Since it is at the brink of being a problem building, IEQ dissatisfaction more than 30% should be avoided.

Conclusions

A new indoor environmental quality comfort model is proposed which is aimed to reflect the total subjective responses of indoor occupants. Coupled with the relational model to the corresponding key physical parameters, the new indoor environmental quality comfort equation is able to predict the total indoor environmental comfort level.

The work in this paper shows that there is a significant correlation of the total environmental dissatisfaction to the individual indoor environmental qualifiers, namely, thermal comfort, indoor air quality comfort and aural comfort. Both pieces of work imply that in general, people response more to a total sensation rather than to a specific indoor environmental qualifier. In real life practice, any building operator may not be able to control the set point for each indoor environmental qualifier precisely, this new IEQ equation provides the flexibility for the operator to meet the indoor users' expectation of their total sensation of IEQ comfort. The suggestion of meeting the expectation for component variables by individual control can be viewed as a special case of the total sensation where each component to set to the highest possible level of comfort.

As the new indoor environmental quality comfort equation provides a handy tool for building developers and designers to make investment plans. An average job with a reasonable provision of resources and ethical job, a 20% PDIEQ is expected. A grade A building would require a higher quality of material, equipment and skill in design, operation and maintenance for 10% PDIEQ. Since this new IEQ index integrates the basic indoor environmental qualifiers, it gives an opportunity to developers an approach to trade off.

References

1. Mui, K.W., Chan, W.T. and Burnett, John. (1999). The Use of Indoor Environmental Quality Logger for Hong Kong Building Environmental Assessment in Office Buildings. In: *Proceeding of International Conference on Urban Pollution Control Technology: Urban Pollution Control Technology*, Edited by Poon, C.S. and Li, X.Z. Department of Civil and Structural Engineering, The Hong Kong Polytechnic University, 615-622.

2. Goldman, R.F. (1999). Extrapolating ASHRAE's Comfort Model. *HVAC&R Research*, **5** (3), 189-194.

3. Haghighat, F. and Donnini, G. (1999). Impact of Psycho-Social Factors on Perception of the Indoor Air Environment Studies in 12 Office Buildings. *Building and Environment*, **34**, 479-503.

4. Clausen, G., Carrick, L., Fanger, P.O., Kim, S.W., Poulsen, T. and Rindel, J.H. (1993). A Comparative Study of Discomfort Caused by Indoor Air Pollution, Thermal Load and Noise. *Indoor Air*, **3**, 255-262.

5. Fanger, P.O. (1988). Introduction of the olf and the decipol units to quantity air pollution perceived by human indoors and outdoors. *Energy and Buildings*, **12**, 1-6.

6. Chan, W.T., Burnett, J., de Dear, R.J. and Ng, C.H. (1998). A Large-scale Survey of Thermal Comfort in Office Premises in Hong Kong. *ASHRAE Technical Data Bulletin*, **14** (1).

7. Chung, T.M. and Burnett, J. (1996). A Field Survey of Lighting Environments in Offices Premises in Hong Kong. Indoor Environmental Research Group, Department of Building Services Engineering. The Hong Kong Polytechnic University.

8. Tang, S.K., Burnett, J. and Poon, C.M. (1996) Aural Environment in Air-conditioned Open Plan Offices. *Building Services Engineering Research and Technology*, **17** (2).

9. ASHRAE (1992). *Thermal environmental conditions for human occupancy*, Atlanta: American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc., 1992. (ANSI/ASHRAE Standard 55-1992)

10. Fanger, P.O. (1998) Ventilation for Buildings – Design Criteria for the Indoor Environment. European Concerted Action. Technical Committee CEN/TC 156. Brussels.

11. Dan, I. (1993). Total Environmental Quality. *ASHRAE Transactions*, 960-967.

12. Wallace, L.A., Nelson, C.J. and Dunteman, G. (1991). Workplace Characteristics Associated with Health and Comfort Concerns in Three Office Buildings in Washington, DC. *Healthy Buildings*, American Society of Heating, refrigerating and Air-Conditioning engineers, Inc., 56-59.

13. Gan, G. and Croome, D.J. (1994). Thermal Comfort Models Based on Field Measurements. *ASHRAE Transactions*, NO-94-6-3, 782-794.